

NETZOOM™



Visual Data Center Infrastructure Management + Service Desk

Away From Your Data Center? You Can See It!



Inventory and Visual Asset Management



Cables and Connectivity Management



Operational Efficiency and Visual Change Management



IT Business Services



Monitoring and Alerts Management



Capacity Analysis and Optimization



IT Service Management



Compliance & Risk Management

Data Center professionals use NetZoom Enterprise with integrated Service Desk to model, monitor, manage, access and control Physical, Virtual and Cloud IT Infrastructure, along with Facility Infrastructure to create the source of truth with an emphasis on capacity analysis and optimizing power, space and cooling to achieve increased server utilization, virtualization and power density to attain desired power usage effectiveness (PUE) and other optimization goals.

NetZoom Enterprise implements the latest technology to fulfill current and future requirements of all Data Centers including Co-lo, Government, Manufacturing, Financial, Medical, Education and other industries.

NetZoom Enterprise is an intuitive, hardware agnostic DCIM platform.

NetZoom is an all-in-one solution with built-in Service Desk, Report Builder, Workflow designer and many out-of-box connectors to integrate with popular applications such as ServiceNow and Remedy. It is easy-to-use, fully scalable and comprehensive, enabling major cost reductions, improved efficiencies, increased productivity and a better overall user experience.

On-Premises | SAAS | Hosted





Inventory and Visual Asset Management

DISCOVER, MODEL, MANAGE AND VISUALIZE EVERY DATA CENTER ASSET

Complete Lifecycle Management of Every Asset - Complete Asset Lifecycle from dock to decommission of Physical, Virtual and Cloud and Facility Infrastructure; Visually Model and Manage in 2D & 3D: buildings, floors and rooms, racks, power, devices, cables, IP cameras, sensors and more.

Extensible and Real-time Asset Information - Devices have pre-populated manufacturer properties, device location properties, computed capacity metrics and real-time data from multiple sources including Intel DCM, RFID, Hypervisors, Cloud services, and CMDB integration. Extend properties using Entity Manager.

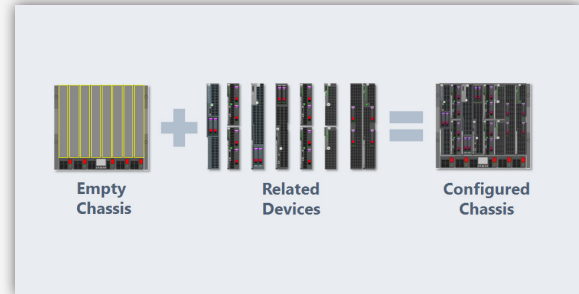
Search, Locate and Filter Assets - Users can locate and navigate to devices deployed anywhere in the data center by searching identifying information like AssetID, Serial Number, Service Tag, RFID Tag, IP address or user defined fields. Use customizable filters to view the data center by Ownership, Management Unit, Project, Capacity, Entity, Device or Tenant.

Asset Ownership and Tenants - Manage Data Center assets by Tenants, Departments, Management Teams or Projects

Inventory Stores - Organize and Track the complete asset inventory by: Installed inventory, Decommissioned inventory, New inventory and Used/spare inventory.

Visio Diagramming - No more manual drag & drop. Auto-diagram Rack elevations, devices and rows of racks in Microsoft Visio.

Device Builds & Rack Builds - Visually create, reuse and manage standard configured builds (racks, row of racks, enclosures/blade servers, switches, servers, etc.) for quick inventory provisioning.



Device Library - Library includes over 200,000 Device models from over 5000 hardware manufacturers. Library is hosted in the cloud for 24x7 availability and updated every week.

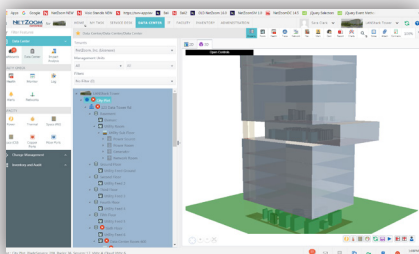
Deploy New Models Faster - Deploy assets using preconfigured chassis and SmartBuilds, inventory stores and the Placement Wizard which locates optimal locations based on criteria.

Zero RU Assets - Track and manage every Data Center asset, even if an asset does not occupy floor space or rack units.

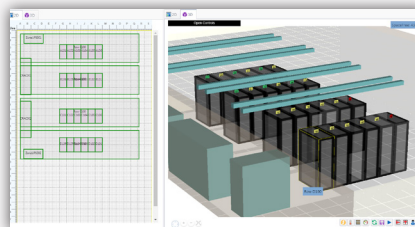
Asset Discovery - Agentless discovery includes the following protocols: WMI, TCP, IPMI, SNMP, SSH, WSMAN, RDP, VNC, and Redfish.

ADVANCED VISUALIZATION

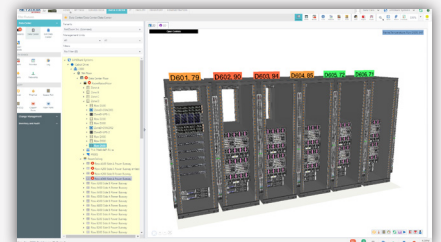
Model Data Center - NetZoom lets users model the entire data center from the site, through the building, floors, rooms, rows, racks, devices all the way to the ports.



Data Center Visualization - Visit the data center from your desk. See all levels of the data center site, building, room, locations, and rows all the way to the racks in 2D or 3D with full detail.



Capacity Overlays - view the device properties, space, power, temperature and other metrics along with the devices and racks.



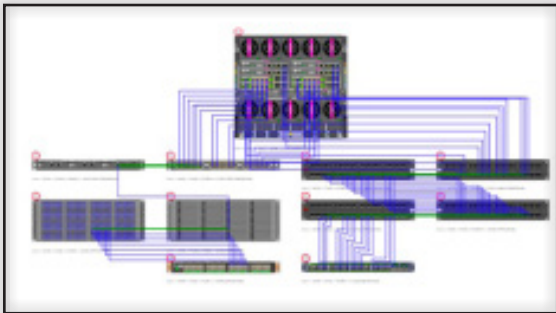


Cables and Connectivity Management

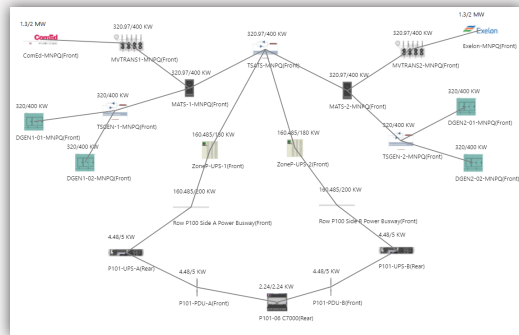
VISUALLY MANAGE POWER AND NETWORK CONNECTIVITY, PATCH PANELS AND CABLES

Port-to-Port Connectivity and Cabling - Model and Manage end-to-end port-level power and network connectivity, copper and fiber cabling, fiber trunking and specialty cables like MPO to LC breakout cables.

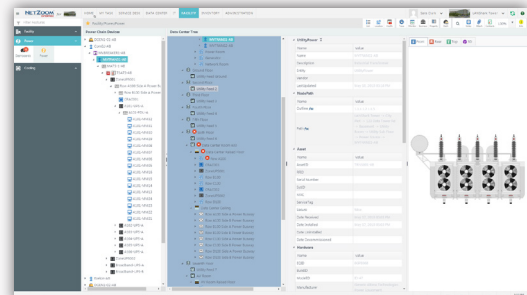
Connectivity Trace - Visualize end-to-end copper and fiber connectivity through every patch panel.



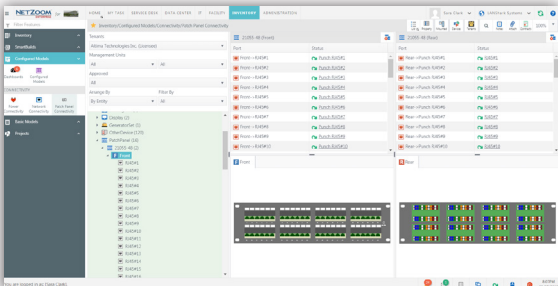
Power Connectivity - Use the Visual Power Chain tracing to view the review power hierarchy including load and capacity at each Power socket to balance power, validate power connectivity and identify single-points-of-failure.



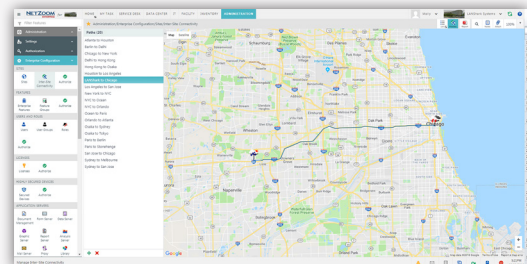
Facility Power - View all levels of facility in power hierarchy.



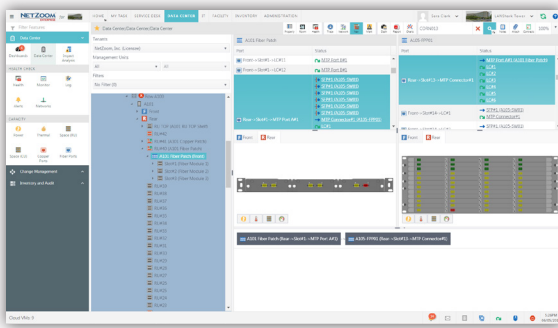
Patch Panel Management - Use the Patch Panel Library and manage the internal connectivity.



Inter-Site Connectivity - Model, visualize and manage connectivity route segments between your geographically disparate data centers or branch office data centers.



MPO Connectors and Hydra Cables - Model and Manage network connectivity through MPO connectors; MPO to LC breakout cables, devices with built-in MPO ports and intermediate hops.



Planned Connectivity - Visually model port-to-port power and network connectivity in change management process. Select cables and automatically generate detailed connectivity instructions.

Connectivity Reports - Generate detailed connectivity reports with automatically calculated cable lengths.

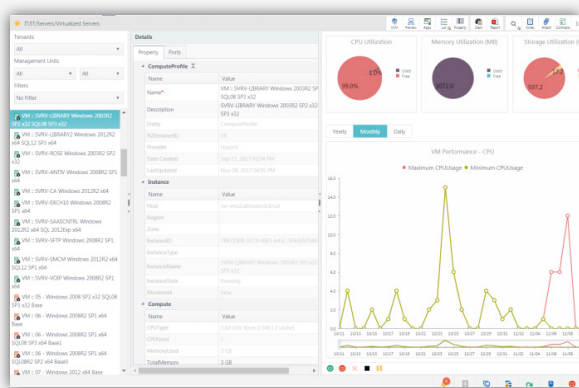


Operational Efficiency and Visual Change Management

CHANGE MANAGEMENT, INFORMATION CONSOLIDATION, AUTOMATION, ANALYTICS AND REPORTING

Single Pane of Glass - NetZoom integrates directly with IT and Facility devices and other installed applications in the data center including CMDBs, hypervisors, cloud compute resources, environmental sensors, RFID systems, Intel DCM and intelligent PDUs, to provide a single interface to manage the entire data center.

Discovery & Monitoring - Discover and monitor physical, virtual, and cloud servers, their platform and applications and control Hosts and Virtual Machines across Hypervisors and cloud providers.



Integrated Document Management System and Contract Management - Centrally store, search, retrieve & manage all important documents like warranties, service contracts, purchase/lease documents, user guides, notes, data files, pictures, and binary files with any entity.

Automated Report Delivery - Schedule auto delivery of selected reports.

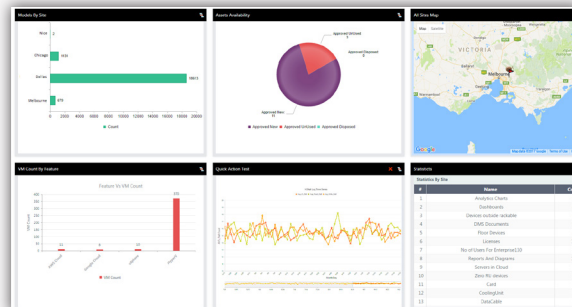
CHANGE MANAGEMENT

NetZoom offers many levels of Change Management.

Planned Change Management - Customizable Change Request Workflows to visually plan Moves, Adds and Changes and Transfers with built-in conflict checking, create work orders and control the entire request-to-change process through review, approval, and execution.

Work Order	
#	Instruction
1	Add device: [SH-100] from [Inventory] to [Row 1].
2	Add device: [SH-100] from [Inventory] to [Floor Region].
2.1	Dual power sources required
3	Add device: [VOL-0600] from [Inventory] and mount [Front] of device: [VOL-0600] from [Inventory] on [Front] of Rack: [SH-100] at [R]#29.
4	Add device: [VOL-0600] from [Inventory] and mount [Front] of device: [VOL-0600] from [Inventory] on [Front] of Rack: [SH-100] at [R]#29.
4.1	Remember to use CAT6 cables for this rack
5	Add device: [I430] from [Inventory] and mount [Front] of device: [I430] from [Inventory] on [Front] of Rack: [SH-100] at [R]#29.
6	Add device: [WS-C2800] from [Inventory] and mount [Front] of device: [WS-C2800] from [Inventory] on [Front] of Rack: [SH-100] at [R]#29.
7	Add device: [C220] from [Inventory] and mount [Front] of device: [C220] from [Inventory] on [Front] of Rack: [SH-100] at [R]#21.
8	Add device: [OR61] from [Inventory] and mount [Front] of device: [OR61] from [Inventory] on [Front] of Rack: [SH-100] at [R]#17.

Customizable Reports, Analytics & Dashboards - Hundreds of ready-to-use and fully customizable Analytics and PDF reports, barcodes and labels, and dashboards for any purpose. Integrated Report Builder and Dashboard Designer.



Impact Analysis - Analyze downstream device dependency and business service map for a device or critical alert.

Task-based UI - Customizable user workspace simplifies user task navigation to improve efficiency.

Automate Routine Tasks - Improve operations with automated database optimization, backup and restore, and diagnostics.

Troubleshooting - Triage device properties, incident management, health history, maintenance log, and connectivity traces to troubleshoot incidents and perform root cause analysis

Filters & Ownership - Manage by scope, use customizable filters to view the data center by Ownership, Management Unit, Project, Capacity, Entity, Device or Tenant.

NetZoom Mobile App - Stay connected with your data center while away from the office.

Quick Changes - Authorized users can "Check Out" a data center node to perform a Quick Change without using the Change Management Approval process.

Unplanned Change for What If Analysis - Use the Unplanned change to perform What If Analysis prior to initiating a change.

Tech Refresh Wizard - Tech Refresh Wizard simplifies bulk replacement of devices with new models during a Technical Refresh cycle.

Best Fit Placement Wizard - Best Fit assists with optimal placement based on various criteria like available rack space, power, thermal, and ports.

CMDB Initiated Change - NetZoom integrates with ServiceNow and Remedy for IT change management service requests.



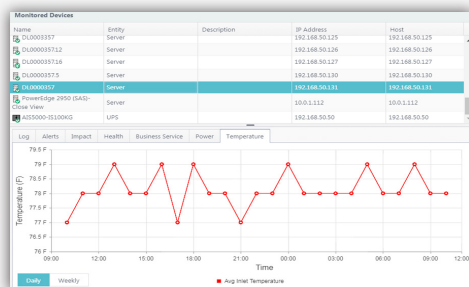
Monitoring and Alerts Management

PROACTIVELY MONITOR YOUR DATA CENTER AND MANAGE ALERTS

Command Center - Create a Command Center to monitor critical data center information using real-time dashboards for environmental conditions, CPU utilization, device health, power consumption, other capacity metrics and physical security with IP cameras.



Monitoring - Real-Time Monitoring of servers, blade servers and Intelligent PDUs for capacity metrics, health conditions and alerts.

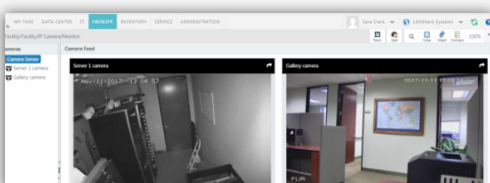


Device Health Check - Monitor the health of devices and device maintenance history to assess potential health risks in the Data Center.

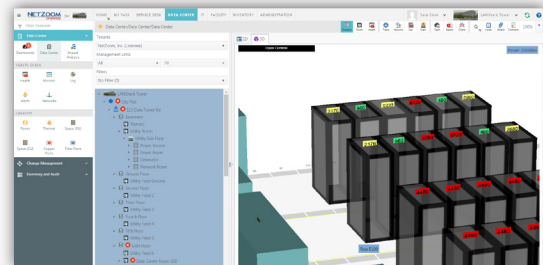
Log	Alerts	Impact	Health	Business Service	Power	Temperature
Battery	Fan	Memory	Power Supply	Storage	Temperature	Voltage
OK	OK	OK	OK	OK	OK	OK

Proactively Monitor - Monitor for Open Ports, Unauthorized Ghost IP addresses, unauthorized software and single points of failure.

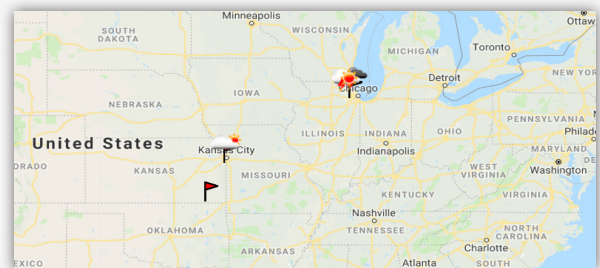
IP Cameras - Enhance data center physical security by monitoring IP cameras. View live streaming and record video from multiple cameras simultaneously.



Interactive 3D View - Real-time data can be viewed interactively in 3D. Choose the real-time capacity values to display Power, Temperature, Space or other information.



Weather Information - Out-of-box integration with Google maps and weather service providers keeps users aware of weather conditions near the data centers.



ALERTS MANAGEMENT

Alert Notification in Real-time - One click actionable critical alerts are generated and delivered online by SMS or email and managed by customizable alert management workflow driven by alert policies and auto-escalation rules.

Threshold-Based Alerts - NetZoom offers threshold-based notification for critical metrics such as Power, Thermal, Weight and Space to monitor and analyze capacity consumption data by data center site, floor, room, rack or enclosures. Threshold violations generate real-time, actionable alerts.

Alerts	Severity	Status	By	Alert
10/10/17 3:02PM	Critical	Open (Level 0)	Sara Clark	[A103-SVR02] Health Alert (Temperature High) on [A103-SVR02] NodePath: [LANShark Systems -> Cabot Drive -> 2300 -> 5th Floor -> Data Center Floor -> RoomRaisedFloor -> Zone A -> Row A100 -> A103 -> A103-SVR02]
10/10/17 3:01PM	High	Open (Level 0)	Sara Clark	[A103-SVR02] Health Alert (Temperature High) on [A103-SVR02] NodePath: [LANShark Systems -> Cabot Drive -> 2300 -> 5th Floor -> Data Center Floor -> RoomRaisedFloor -> Zone A -> Row A100 -> A103 -> A103-SVR02]

Third Party Alerts - Alerts generated in third party applications are managed, delivered, escalated and attended in NetZoom.



Capacity Analysis and Optimization

MAINTAIN YOUR CAPACITY AT AN OPTIMAL LEVEL

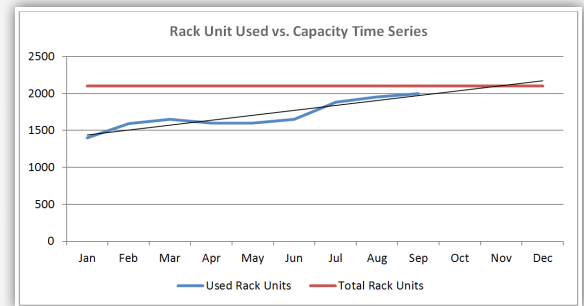
Executive Dashboards - Dozens of ready to use reports to stay updated at all times: what do we have? How are we doing? Where are we heading? What can be optimized?

Capacity - Know and Manage the Capacity: Power, Thermal, Space, Weight, Ports and Compute.

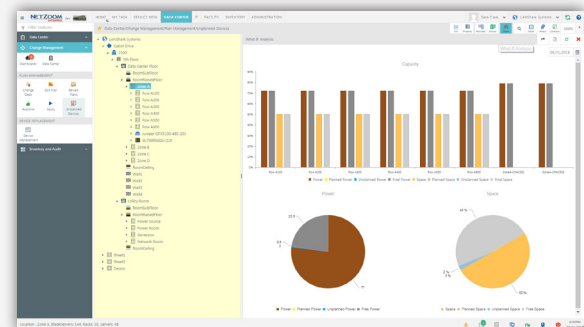
Efficiency - Know data center efficiency metrics: space utilization, power utilization, cooling utilization, port utilization, % of virtualization.



Understand Capacity Utilization - Compare provisioned capacity to actual values and trends.



What If Analysis - Evaluate the impact of planned and unplanned changes to the physical infrastructure with What If Analysis.



CAPACITY OPTIMIZATION

Reducing Over-Provisioning - Monitoring real-time power usage over time vs. provisioned power to reduce over-provisioning and redistribute provisioned power.

Power Capping - Use power capping to reduce consumption and increase availability in case of CRAC or power device failure.

Identifying Ghost Servers - NetZoom identifies Ghost Servers that are consuming resources.

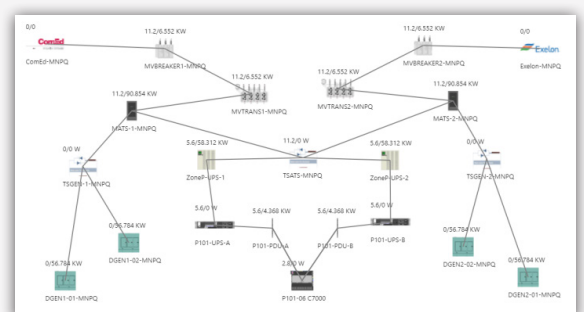
Reduce Stranded Capacity - Use the Composite Capacity report to help identify and balance Power, Space and Temperature to reclaim stranded capacity.



Increase Availability - Conserving Power & Cooling Capacity through power capping profiles for increased availability.

Increase Virtualization - Identify Underutilized Servers that can be virtualized.

Balance the Power Equation - Visual Power Chain to balance Power Capacity with Power Load at PDU level.





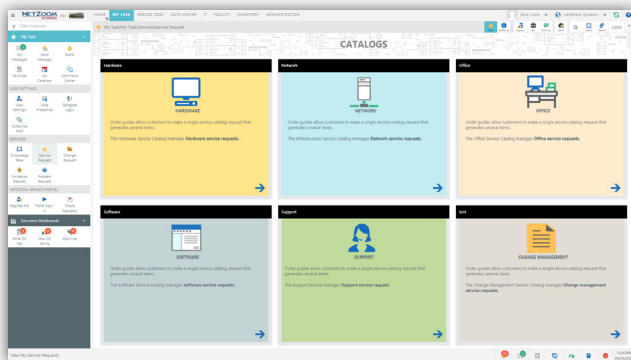
IT Service Management

DATA CENTER SERVICE REQUEST & INCIDENT MANAGEMENT

Create services to match your unique processes and workflow.

Self-Serve & Knowledgebase - Users can quickly find answers to issues by searching the knowledge base articles, tutorials, workarounds, FAQs and common requests.

Service Catalog – Define services for Service Catalog and Efficiently Manage the Service Requests.



Service Request – Users request services provided by the data center.

VM Service Form

Requested By: NetZoom Enterprise Demo

Additional Users To Be Notified: [Dropdown]

Request Name: Need new VM

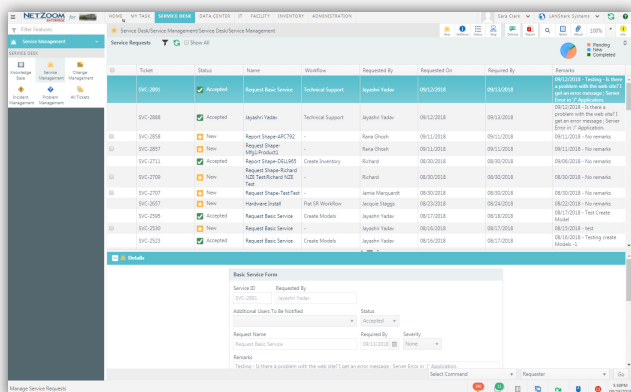
Remarks: I need a new VM for my NetZoom Enterprise install

Workflow: Select a Workflow

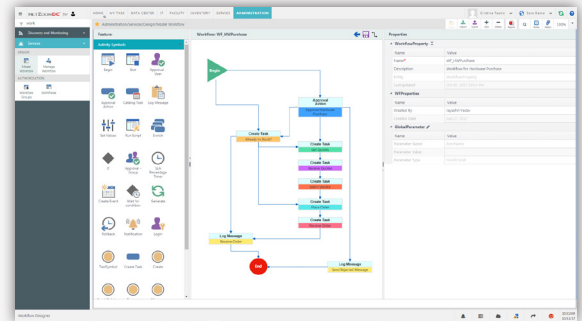
VM Configuration: Windows Server 2016 with SQL Server 2012 SP4. Need 4 Cores and 16GB of RAM. See attached Checklist for full VM requirements.

Update Cancel

Service Desk - Manage Service Requests from all users.



Workflow For Every Service - Graphical Workflow Designer enables users to design service workflows for every service offered by the data center. Workflows automate services requests and task-oriented processes while following authorization, policy, escalation and delivery rules.



Events & Incident Management – NetZoom generates tickets for Incident workflow, notifies stakeholders, auto routes tickets using escalation criteria and manages ticket resolution.

My Task/My Service Requests/Service Request

Incident Service Form

Requested By: Sara Clark

Request Date: 11/11/17

Status: Open

Required Date: 11/11/17

Last updated: 11/11/17

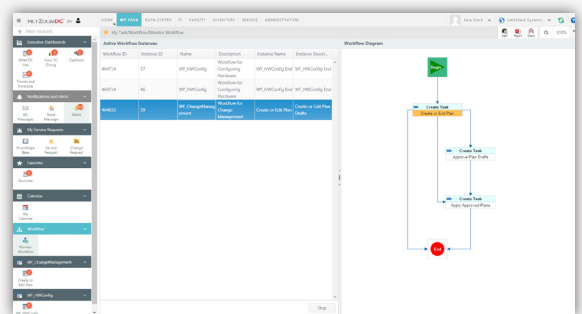
Request Name: Exchange Logon

Service ID: [Dropdown]

Severity: Medium

Description: Please see what is happening. I am unable to log onto the Exchange server. My credentials were fine yesterday.

Change Request Management - User customizable Change Request Workflows to efficiently manage planned infrastructure changes through the entire change process while maintaining the change audit trail for compliance.



Service Alerts - Manage Data Center Service Alerts triggered by application performance, maintenance and SLA requirements.

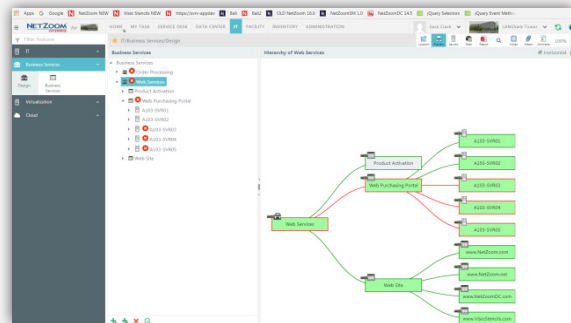


IT Business Services

BRIDGE THE GAP BETWEEN IT AND THE BUSINESS

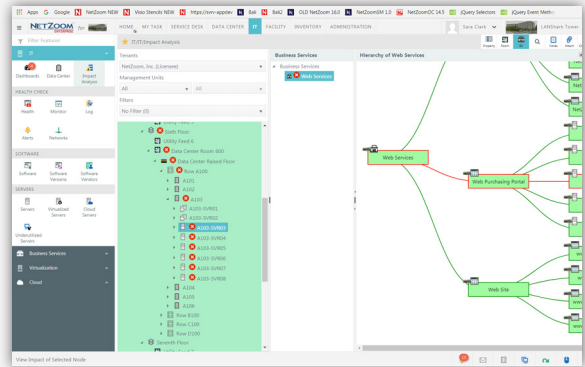
Manage and Monitor IT Business Services – Model Business Services offered by the data center.

Business Services Dependency Map - View the relationships and dependencies between business services and IT components such as servers, networks, storage, cloud resources and related services that deliver the Business Service.



Service Alerts - Manage Data Center Service Alerts triggered by application performance, maintenance and SLA requirements.

Business Service Impact Analysis - Predict impact to Business Services on demand using Impact Analysis.



Compliance & Risk Management

ENSURE COMPLIANCE INITIATIVES & IDENTIFY FAILURE RISKS

Compliance Regulations - Meet compliance regulations by using NetZoom's:

- Authentication, Role-Based Authorization and Security Policies
- Compliance Reporting
- DCOI Dashboard
- Infrastructure Inventory Collection and Audit
- Forensic and Change Logs
- Notes & Documentation

Proactive Risk Management - proactively identify potential risks:

- Identify Open Ports
- Identify Unauthorized IP Addresses
- Identify Unauthorized Software
- Identify Single Points of Failure
- Identify Failing Hardware using Health Check

Physical Environment and Security - Minimize physical security risks to the data center by using:

- Weather Map
- IP Cameras
- Environmental Sensors
- Asset and Cable Labeling
- Asset Tracking and Locating

Financial Risk - Reduce the Financial Risk by knowing and managing:

- Warranty alerts
- Residual value of Assets
- Value of spare inventory
- Residual value of Warranty
- Assets out of warranty and EOL

Out-of-Box Integration and Development Tools

NetZoom integrates out-of-box with popular applications including hypervisors such as vSphere and Hyper-V, cloud service providers such as Google Cloud, Amazon Web Services, and Microsoft Azure, CMDBs such as ServiceNow and Remedy, Intel DCM, RFID, sensor technology, and PDUs and UPSs. Visit NetZoom.com for an up-to-date list of integrations.

NetZoom offers built-in application enhancement tools like Report Builder, Workflow Designer and Entity Manager that enable further customization. The NetZoom REST API set is available in the NetZoom SDK to facilitate integration with other data center applications.

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